

AMITASH NANDA

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EDUCATION

University of California San Diego , San Diego (CA), USA	September 2023 – June 2027
Doctor of Philosophy (Ph.D.) in Electrical and Computer Engineering (Machine Learning and Data Science)	GPA: 3.5
Research: Distributed ML, Generative AI, Computer Vision, DNN/LLM Optimization, AI/ML Systems, Applied Data Science	
University of California San Diego , San Diego (CA), USA	September 2021 – June 2023
Master of Science (M.S.) in Electrical and Computer Engineering (Intelligent Systems, Robotics and Control) [Thesis]	GPA: 3.5
Odisha University of Technology and Research , Bhubaneswar (OD), India	August 2015 – May 2019
Bachelor of Technology (B.Tech.) in Instrumentation and Electronics Engineering	GPA: 4.0

TECHNICAL SKILLS

Languages/Databases	Python, C++/CUDA, JavaScript, Bash/Shell, SQL (MySQL, PostgreSQL), GraphQL, MongoDB, Redis
Frameworks	PyTorch, TensorFlow, Keras, Hugging Face Transformers, DeepSpeed, Spark, ONNX, YOLO, Flask
Libraries	Scikit-learn, Pandas, NumPy, Dask, OpenCV, NLTK, OpenMP, TensorRT, cuDNN, SHAP, Captum
Tools	Anaconda, Jupyter, LaTeX, Git/GitHub, Jenkins, Docker, Kubernetes, Slurm, Grafana, MLflow, MLPerf
Cloud/Technologies	AWS (SageMaker, EC2, Lambda), Azure ML, GCP, HPC Systems (NERSC Perlmutter), LangChain

EXPERIENCE

Graduate Student Research Assistant NERSC (LBNL) , Berkeley (CA), USA	June 2025 – September 2025
Data and Analytics team: Led the development of a foundation AI model for protein understanding and design on HPC. - Engineered a novel guided-generation framework integrating ESM-3 and custom scoring function for protein stability.	
Graduate Student Research Assistant NERSC (LBNL) , Berkeley (CA), USA	June 2024 – September 2024
Expanded load-balancing algorithms in AMReX, parallelized a brute-force approach, developed a hybrid SFC-Knapsack and improved SFC bisection strategy using painter's, achieved faster run-time, and 99.8–99.9% balanced efficiency at scale.	
Teaching Assistant CSE, Data Science, ECE (UCSD) , San Diego (CA), USA	March 2024 – Present
- Managed 60 students across six groups for the Data Science Capstone Project, guided them through ML model optimization, NLP, Gen-AI, and Graph ML. Assisted as a TA in the Advanced Data Structures course (CSE 100). - Facilitated discussions, office hours, and assessments, supported group projects and provided guidance for code checkpoints.	
Graduate Student Researcher Boolean Lab , San Diego (CA), USA	September 2021 – Present
Computational Models: Analyzing large-scale biological datasets to identify Boolean relationships between genes and developing computational methods to accelerate the use of AI in pathology and drug discovery. AI Integration and Model Optimization: Building multimodal approaches, foundation models, and fine-tuning strategies for vision and language-related tasks to generate biological hypotheses and identify candidate biomarkers.	
Software Engineer Intern – Optimization Group Teradata , San Diego (CA), USA	June 2023 – September 2023
Query Optimization: Leveraged Primary Index (PI) for efficient local aggregation and join capabilities to enhance Teradata's Object File Storage (OFS) system and achieved 20% query cost reduction. Optimized data scenarios for 1B rows. Framework: Developed an automated framework for data and query generation, simulated good and bad cases for DBQL performance benchmarking. Generated 512 objects with 700M rows (good case) and 233 objects with 48M rows (bad case).	
Research Intern – Technology Innovation Office Teradata , San Diego (CA), USA	June 2022 – September 2022
Performance Optimization: Integrated predictive models to forecast platform configuration resource usage, analyzed telemetry data from the Teradata Telemetry Collection Agent (TCA), and improved operational efficiency. Automation: Designed and implemented an end-to-end automated pipeline, integrated data retrieval from TCA using REST API, performed data preprocessing, built predictive models, and streamlined platform monitoring and forecasting.	
Software Engineer II Accenture , Bengaluru (KA), India	June 2019 – August 2021
Application Development: Designed and built a novel robot software testing application (chaosRobo) utilizing chaos engineering principles to simulate real-world scenarios. Led the first-phase prototype development team. Testing Framework: Contributed to backend GUI design, integrated Gazebo-ROS functionalities, and successfully deployed the system on AWS S3 and RoboMaker, enhancing robot simulation and testing efficiency. Performance Optimization: Implemented machine learning-based analytics in chaosRobo to assess system robustness, reducing failure detection time by 30% and improving overall testing accuracy by 20–25%. Bedrock Automation Asset Development: Automated end-to-end user stories for Airbus PPR scenarios using image-based and text-based recognition. Devised a new testing methodology and integrated it into the 3DEXPERIENCE software.	

PROJECTS

- ESM-3 Guided-Generation Based Protein Engineering | NERSC (LBNL) [GitHub] [Package]** June 2025 – Present
- Developed a guided-generation framework integrating **ESM-3** generative model with **FoldX** scoring function (**ddG** scores).
 - Enhanced framework capabilities to refine sequences for thermostability and binding affinity, reducing time for each generation step by **75–80%** and achieving **8x** faster performance leveraging **HPC-AI** workflow. Published a **PyPI** package.
- Dynamic Load Balancing Algorithms in AMReX | NERSC (LBNL) [GitHub]** June 2024 – March 2025
- Introduced a novel hybrid load-balancing algorithm combining SFC and Knapsack, improved SFC bisection and Knapsack using **painter's** algorithm, optimized dynamic load-balancing in AMReX framework on NERSC Perlmutter supercomputer.
 - Statistically analyzed the efficiency and runtime of these algorithms for sizes up to **512** ranks. Knapsack combined with painter's achieved highest efficiency ranging **99.8–99.9%**, while SFC with painter's algorithm demonstrated faster runtime.
- CHAI-KTQ: Framework for Scalable LLMs & Efficient Inference | Boolean Lab [GitHub]** August 2024 – Present
- CHAI-Quant performs sensitivity-aware mixed-precision quantization on clustered attention heads, reducing KV cache size by up to **57.8%**. CHAI-Target performs sensitivity-aware fine-tuning improving accuracy **22–24%** over baseline CHAI.
 - CHAI-KD implements an efficient teacher-student knowledge transfer, achieving inference speed of **3x (3,000 inf/sec)**.
- CPTQuant: Mixed Precision Post-Training Quantization for LLMs | Boolean Lab [GitHub]** May 2024 – Present
- CMPQ** adjusts precision using canonical correlation analysis across layers. **PMPQ** applies layer-wise precision optimization based on sparsity sensitivity. **TDMPQ** leverages Taylor decomposition to adapt precision to input perturbations.
 - CPTQuant** achieved up to **4x** compression and a **2x** efficiency over Hugging Face **FP16**. **PMPQ** delivered **11%** higher compression for classification tasks, while **TDMPQ** achieved a **30%** greater compression ratio for language modeling tasks.
- Communication Efficient Asynchronous Peer-to-Peer Federated LLMs | Boolean Lab [GitHub]** December 2023 – April 2024
- Developed a secure, efficient, and privacy-preserving federated learning framework in a decentralized setting, achieving **5x** lower latency and **13%** higher accuracy in serverless environments.
 - Observed a **76%** reduction in information passing time for asynchronous methods, while PageRank proved most effective at removing anomalous nodes, boosting the global federated LLM model's performance.
- Expression Gradient of Cancer Suppressor Gene using Vision-AI | Boolean Lab [GitHub]** May 2022 – May 2023
- Built a novel colon gland instance segmentation model using Mask R-CNN and YOLOv8 to diagnose colon cancer. Mask R-CNN achieved F1/IoU scores of **0.63/0.46** (glands) and **0.59/0.42** (crypts). Annotated first public U-shaped colon dataset.
 - Demonstrated that YOLOv8 outperformed Mask R-CNN with mAP50 scores of **0.937** (glands) and **0.748** (crypts). Predicted **5,000+** U-shaped glands across **25** slides, identifying differentially expressed genes along crypts.

PUBLICATIONS

- Nanda, A.**, Chowdhury, M.K.H., Ross, H., and Gott, K. (2025). Exploring Dynamic Load Balancing Algorithms for Block-Structured Mesh-and-Particle Simulations in AMReX. In *Proceedings of the ACM PEARC*. DOI: **10.1145/3708035.3736022**
- Nanda, A.**, Baliya, S.B., and Sahoo, D. (2025). CHAI-KTQ: A Novel Framework for Scalable LLMs and Efficient Inference. Manuscript in preparation.
- Behroozikhah, M., **Nanda, A.**, Khandelwal, S., and Sahoo, D. (2025). Expression Gradient of Cancer Suppressor Gene Found in Colon Crypt using Vision-AI. Manuscript in preparation.
- Nanda, A.**, Haque, H.M.Z., Zhou, X., and Sahoo, D. (2025). GeneST: A Deep Learning Framework to Identify Spatially Distributed Genes for Translational Health Impact. Manuscript under review.
- Nanda, A.**, Baliya, S.B., and Sahoo, D. (2025). FedNAMs: Performing Interpretability Analysis in Federated Learning Context. Preprint. arXiv. DOI: **10.48550/arXiv.2506.17466**
- Nanda, A.**, Baliya, S.B., and Sahoo, D. (2024). CPTQuant: Novel Mixed-Precision Post-Training Quantization Techniques for Large Language Models. Preprint. arXiv. DOI: **10.48550/arXiv.2412.03599**
- Baliya, S.B., **Nanda, A.**, and Sahoo, D. (2024). Building Communication Efficient Asynchronous Peer-to-Peer Federated LLMs with Blockchain. In *Proceedings of the AAAI Spring Symposium Series*. DOI: **10.1609/aaais.v3i1.31212**
- Nanda, A.**, Sahoo, D., and Lin, B. (2023). Novel Vision-AI Techniques for Morphological Discovery in Systems Biology. Master's Thesis, University of California San Diego, *Open Access Publication*. URL: **escholarship.org/uc/item/0xh5w6wb**
- Dang, D., **Nanda, A.**, Sahoo, D., and Lin, B. (2023). NeuCASL: From Logic Design to System Simulation of Neuromorphic Engines. Preprint. arXiv. DOI: **10.48550/arXiv.2208.03500**

RECOGNITION / OPEN-SOURCE CONTRIBUTION

- MLCommons (**MLPerf Inference v5.1**) – Benchmarked **ResNet-50** with Apple M1 Pro (10-core CPU, 16-core GPU) using ONNX runtime v1.19.2 with Apple's CoreML/ANE across Offline, SingleStream, and MultiStream. [Website] [GitHub]
- Reviewer for AAAI 2026, ICLR 2025-2026, Poster Judge for Academic Data Science Alliance 2023 (UTSA).
- Selected for **ACM PEARC 2025** student program, served as Poster Judge, facilitated Nvidia's workshop as a lead volunteer.